
USB to EtherCAT Adapter Board V1.0

The domain of the Project:
PCB DESIGN

Team Mentors (and their designation):
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Hardware Design Engineer

Team Members:
Mr. Sangam Bharath Kumar

Period of the project

August 2026 to September 2026



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Declaration

The project titled “USB to EtherCAT Adapter Board V1.0” has been mentored by, organised by SURE Trust, from August 2026 to September 2026, for the benefit of the educated unemployed rural youth for gaining hands-on experience in working on industry relevant projects that would take them closer to the prospective employer. I declare that to the best of my knowledge, the members of the team mentioned below have worked on it successfully and enhanced their practical knowledge in the domain.

Team Members:

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Mentor's Name

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Executive Summary

The project involves the design and development of a **USB-to-EtherCAT Adapter Board V1.0**, intended to provide communication between a USB host system and an EtherCAT network. The board combines an **STM32F411CEUx microcontroller** for USB interfacing and control with the **LAN9252 EtherCAT controller** for handling EtherCAT communication. It also includes dual RJ45 interfaces, an I²C EEPROM for non-volatile data storage, and a 5 V to 3.3 V regulated power supply for the board's electronic components.

The design was developed starting from the circuit schematic, where the major functional blocks and their supporting components were defined. The schematic was then converted into a PCB layout, with attention given to component placement, signal routing, power distribution, grounding, and the connections between the different functional sections. The PCB design was completed using **KiCad**, followed by design verification and 3D visualization of the board.

The completed design brings the USB interface, microcontroller, EtherCAT controller, network connectors, memory, and power supply together on a single PCB. Through this project, the complete PCB design process was explored, from **schematic development and component selection to PCB layout, routing, and design verification**. The project also provided practical experience in designing a communication-oriented board while considering the physical and electrical requirements involved in PCB development.



Introduction

Background and Context of the Project

Industrial automation systems require reliable communication between controllers, computers, and field devices for exchanging data in real time. **EtherCAT (Ethernet for Control Automation Technology)** is a high-speed industrial communication technology designed for efficient and deterministic data exchange. However, connecting an EtherCAT network directly to a computer may require a suitable interface between the host system and the industrial network.

This project focuses on the design of a **USB-to-EtherCAT Adapter Board V1.0**, which provides an interface between a USB host and an EtherCAT network. The board combines a microcontroller with the EtherCAT controller, along with the required USB, RJ45, memory, clock, and power-supply circuits. The project provides a practical implementation of these functional blocks on a single PCB and demonstrates the process of developing a communication-oriented electronic board from schematic to PCB layout.

Problem statement and goals

The objective of this project is to design a **USB-to-EtherCAT Adapter Board V1.0** that integrates the required USB interface, processing unit, EtherCAT controller, network interfaces, memory, and power supply on a single PCB. The design uses an STM32F411CEUx microcontroller together with the LAN9252 EtherCAT controller to establish the required communication interface between the USB and EtherCAT sides.

Scope and limitations

The scope of the project covers schematic design, component selection, PCB layout, component placement, signal routing, power and ground distribution, and design verification using KiCad. The design includes supporting circuits such as the 5 V to 3.3 V voltage regulator, I²C EEPROM, clock circuitry, and RJ45 interfaces.



The project is primarily focused on the PCB design and does not include the development of a complete commercial EtherCAT software system or extensive hardware performance testing.

Practical aspect

The main practical aspect of the project is the integration of multiple communication, processing, memory, and power-management functions into a single PCB. The project involves converting the functional schematic into a physical board while considering component placement, signal routing, grounding, power distribution, and overall layout organization. This provides practical experience in the complete PCB development process, from schematic creation to PCB layout and design verification.



Project Objectives

Project Objectives

The main objective of this project is to design a **USB-to-EtherCAT Adapter Board V1.0** that integrates the required hardware for communication between a USB host and an EtherCAT network. The project aims to develop the complete PCB design while considering the electrical and physical requirements of the different functional blocks.

The specific objectives are to:

- Design the schematic integrating the STM32F411CEUx microcontroller and LAN9252 EtherCAT controller.
- Provide the required USB and dual RJ45 interfaces for host and EtherCAT connectivity.
- Design the 5 V to 3.3 V power supply, clock, reset, EEPROM, and supporting circuits.
- Develop the PCB layout with appropriate component placement, signal routing, power distribution, and grounding.
- Perform design-rule checking (**DRC**) and review the completed layout for possible design issues.
- Generate a 3D representation of the completed PCB to verify component placement and overall board arrangement.

Expected Outcomes and Deliverables

The expected outcome is a complete and organized PCB design that represents the USB-to-EtherCAT adapter at the hardware level. The main deliverables include the **schematic, PCB layout, verified design files, and 3D visualization of the board**. The completed design is intended to provide a suitable foundation for PCB fabrication and further hardware testing.



Methodology:

The design of the PCB was carried out in several stages, starting with understanding the requirements of the USB-to-EtherCAT adapter and identifying the main circuit blocks. The major components, including the STM32F411CEUx microcontroller, LAN9252 EtherCAT controller, USB interface, RJ45 connectors, EEPROM, and power-supply circuit, were selected based on the required functionality. Their datasheets were referred to for pin configurations, operating voltages, recommended connections, and other electrical requirements.

The circuit schematic was then developed using KiCad. The different functional sections were connected with the required supporting components such as decoupling capacitors, resistors, clock circuitry, reset circuitry, and power-filtering components. The completed schematic was checked to ensure that the connections and power requirements were properly defined.

After completing the schematic, suitable footprints were assigned to the components and the design was transferred to the PCB layout. The components were arranged according to their functional sections, with the main ICs and their supporting components kept relatively close to each other. The PCB was then routed using the available copper layers, with attention given to signal connections, power and ground connections, component spacing, and the overall arrangement of the board.

The completed layout was checked using Design Rules Check (DRC) in KiCad to identify possible PCB layout errors. The 3D Viewer was also used to inspect the component placement and overall physical appearance of the board. The Gerber file was generated after completion of the design. Thus, the project covered the complete PCB design process from schematic development and component selection to PCB layout, routing, and design verification.



Tools/Software used:

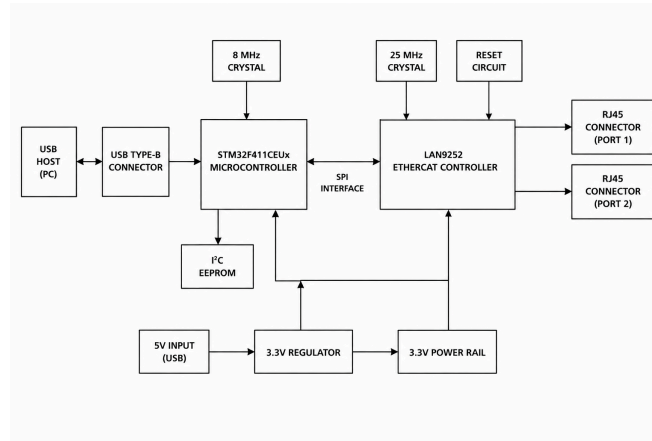
The development of the USB-to-EtherCAT adapter PCB requires tools for schematic design, component selection, PCB layout, verification, and reference of electrical specifications. The following tools and resources were used during the project:



- **KiCad:** Used as the main PCB design software for creating the schematic, assigning component footprints, designing the PCB layout, routing tracks, performing design-rule checks, and generating the 3D visualization of the completed board.
- **KiCad Schematic Editor:** Used to create the circuit diagram and establish the electrical connections between the STM32F411CEUx, LAN9252, USB interface, RJ45 interfaces, EEPROM, power supply, and supporting components.
- **KiCad PCB Editor:** Used for component placement, track routing, vias, copper zones, board outline creation, and overall PCB layout development.
- **KiCad 3D Viewer:** Used to inspect the physical appearance of the PCB and verify the placement and orientation of components before fabrication.
- **Component Datasheets:** Used to obtain important information such as pin configurations, operating voltage, electrical characteristics, recommended component values, and application requirements.
- **Manufacturer Reference Documentation:** Used to understand the recommended implementation and supporting circuitry for major components such as the **LAN9252 EtherCAT controller and STM32F411CEUx microcontroller**.



Project Architecture:



The USB-to-EtherCAT adapter is organized around two main processing and communication devices: the **STM32F411CEUx microcontroller** and the **LAN9252 EtherCAT controller**. The STM32F411 provides the interface on the USB side, while the LAN9252 handles the EtherCAT communication and connects to the external network through the RJ45 interfaces.

The major functional blocks of the system are:

- **USB Interface:** Provides the connection between the host computer and the adapter board and supplies the board with 5 V power.
- **STM32F411CEUx Microcontroller:** Acts as the main microcontroller on the USB side and manages the interface and control functions of the board.
- **LAN9252 EtherCAT Controller:** Handles the EtherCAT communication and provides the interface between the microcontroller and the EtherCAT network.
- **EtherCAT / RJ45 Interface:** Two RJ45 connectors provide the external network connections required for EtherCAT communication.
- **I2C EEPROM:** Provides non-volatile memory for storing required configuration or device-related information.
- **Power Supply:** The incoming 5 V supply is regulated to **3.3 V** to power the required digital circuits on the board.



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- **Clock and Supporting Circuits:** Crystal circuits, reset circuitry, decoupling capacitors, resistors, and other supporting components are provided to ensure proper operation of the major ICs.



Chapter III-Implementation

Schematic Capture:

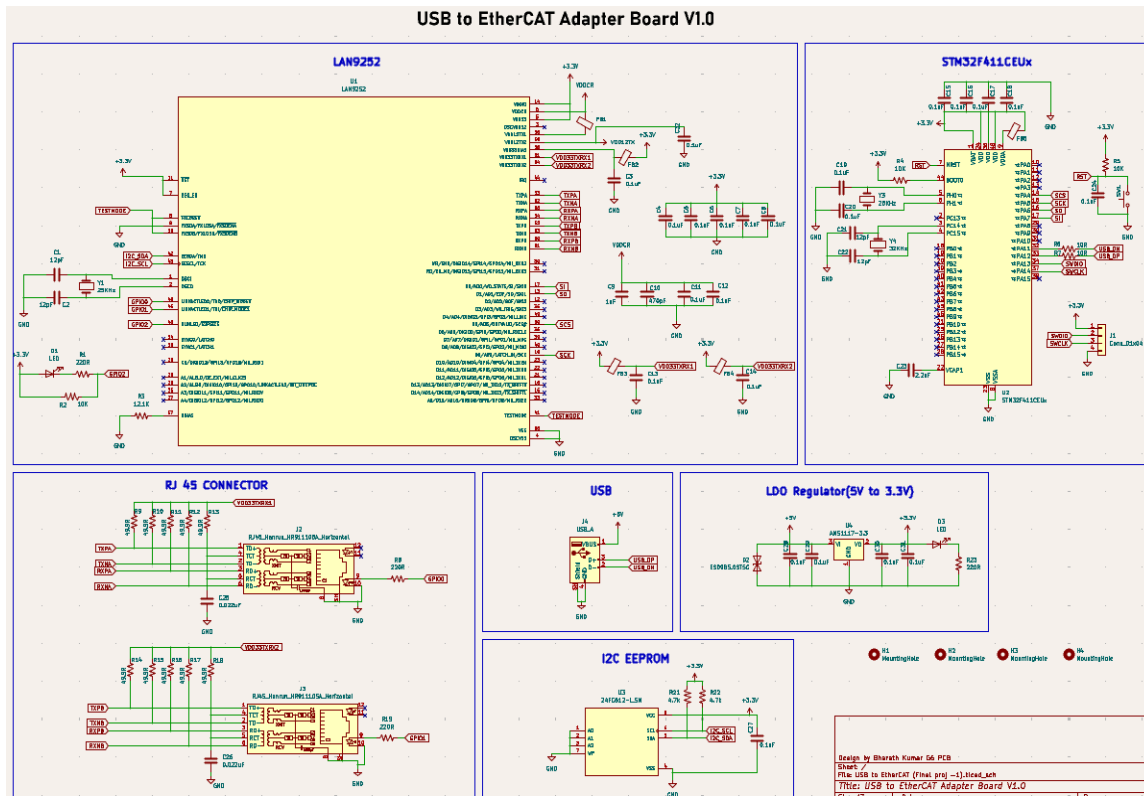


Figure (a)

Figure(a) shows the complete schematic of the USB-to-EtherCAT Adapter Board V1.0. The circuit is divided into several functional sections, including the STM32F411CEUx microcontroller, LAN9252 EtherCAT controller, RJ45 interfaces, USB interface, and supporting circuitry. Each section performs a specific function and is interconnected to form the complete communication and control system.



The diagram shows a USB connector labeled J4 USB_A. The pins are connected as follows:

- Pin 1: +5V
- Pin 3: USB_DP
- Pin 2: USB_DN
- Pin 4: GND
- Shield: GND

The USB Type-A connector provides the connection between the host computer and the adapter board. It is responsible for transferring USB data between the host system and the microcontroller, while also providing the 5 V supply required by the board.

The **D+ and D-** lines are connected to the corresponding USB pins of the STM32F411CEUx for data communication. The **5V VBUS line** is used as the input to the board's power supply section, while the ground connection provides the common reference for the circuit.

[illegible]

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The **LAN9252** is the main EtherCAT controller used in the PCB and forms the central communication block between the STM32F411CEUx microcontroller and the EtherCAT network. It handles the EtherCAT communication and provides the required interfaces for connecting the board to the external network through the two RJ45 ports. The LAN9252 section also includes the supporting clock, reset, power, decoupling, and interface circuitry required for its operation.

Key Specifications:

- **Device:** LAN9252 EtherCAT Controller
- **Function:** EtherCAT slave controller
- **Ethernet ports:** Two, for EtherCAT IN and OUT
- **Host interface:** SPI / digital host interface with the STM32
- **Supply voltage:** 3.3 V
- **Clock:** External crystal/clock circuitry as required by the device
- **Interface:** EtherCAT over Ethernet physical layer
- **Supporting circuitry:** Decoupling capacitors, reset circuitry, clock circuit, and required passive components



STM32F411CEUx Microcontroller

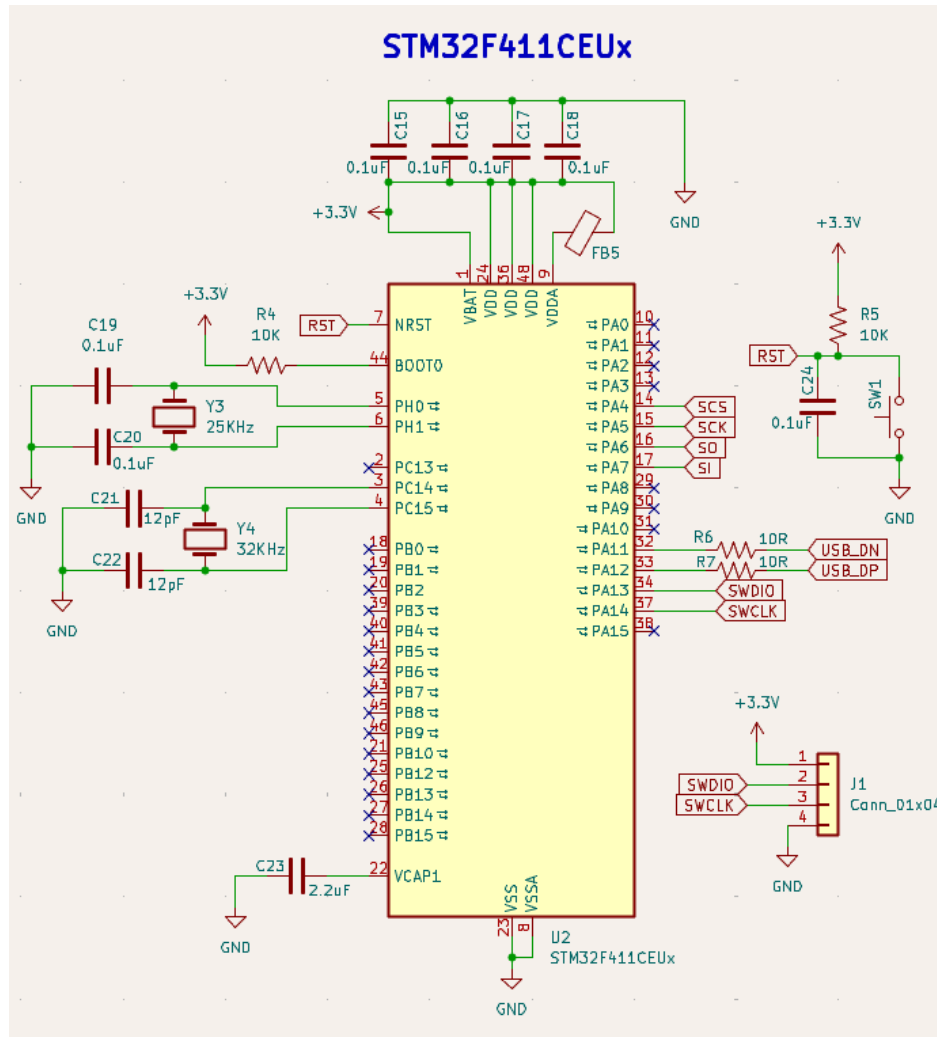


Figure (d)

The **STM32F411CEUx** acts as the main microcontroller on the USB side of the adapter. It interfaces with the USB connection and communicates with the **LAN9252 EtherCAT controller** through the required digital interface. The schematic includes the supporting circuits required for reliable operation of the microcontroller, including power-supply decoupling, clock generation, reset, boot configuration, USB connections, and an SWD programming/debug interface.



Key Specifications:

- **Microcontroller:** STM32F411CEUx
- **Core:** ARM Cortex-M4 with Floating Point Unit (FPU)
- **Maximum operating frequency:** Up to 100 MHz
- **Flash memory:** 512 KB
- **SRAM:** 128 KB
- **Operating supply:** 3.3 V

5 V to 3.3 V LDO Regulator

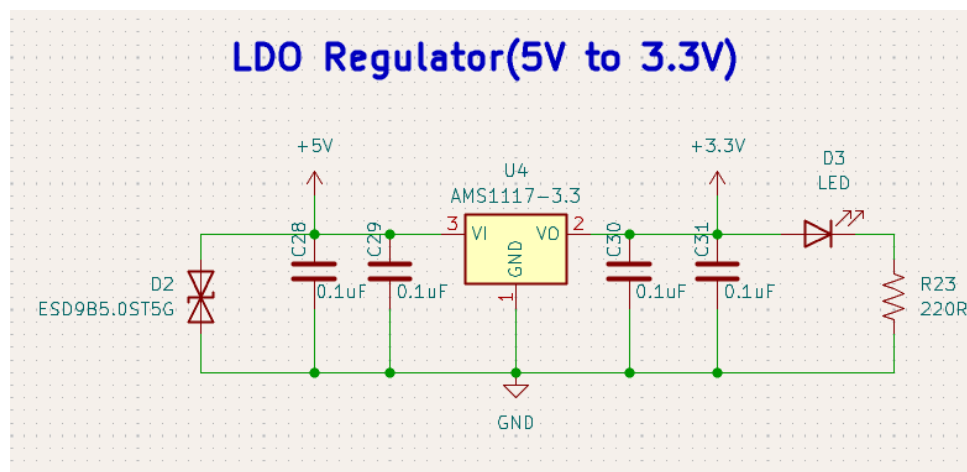


Figure (e)

The 5 V to 3.3 V power supply section is used to generate a stable 3.3 V supply for the digital components on the board. The **AMS1117-3.3** LDO regulator receives the 5 V input and regulates it down to 3.3 V. Input and output capacitors are provided for supply filtering and regulator stability, while an ESD protection device is included on the input side.

A power LED is also provided to indicate the presence of the regulated 3.3 V supply.

Key Specifications:

- **Regulator:** AMS1117-3.3



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- **Input voltage:** 5 V
- **Output voltage:** 3.3 V
- **Input capacitors:** $0.1 \mu\text{F} \times 2$
- **Output capacitors:** $0.1 \mu\text{F} \times 2$
- **Input protection:** ESD protection device (D2)
- **Power indication:** LED (D3)
- **LED current-limiting resistor:** 220Ω

I²C EEPROM

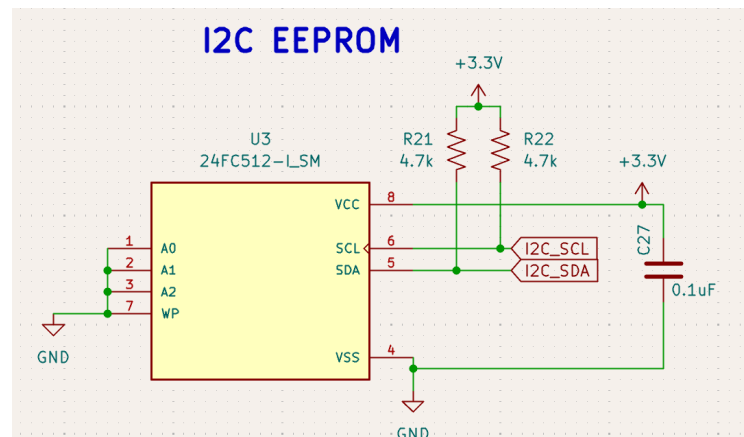


Figure (f)

The **I²C EEPROM** provides non-volatile memory for storing configuration and other required device information. In this design, the **24FC512-I/SM** EEPROM operates using the I²C communication interface, with the **SCL** and **SDA** lines connected to the system through appropriate pull-up resistors. The address pins and write-protect pin are connected to ground in the schematic, defining the device configuration and allowing the required memory operation.

Key Specifications:

- **EEPROM:** 24FC512-I/SM
- **Memory capacity:** 512 Kbit
- **Interface:** I²C
- **Supply voltage:** 3.3 V



- **I²C lines:** SCL and SDA
- **Pull-up resistors:** 4.7 k Ω for SCL and SDA
- **Address pins (A0, A1, A2):** Connected to GND

RJ45 / EtherCAT Interface

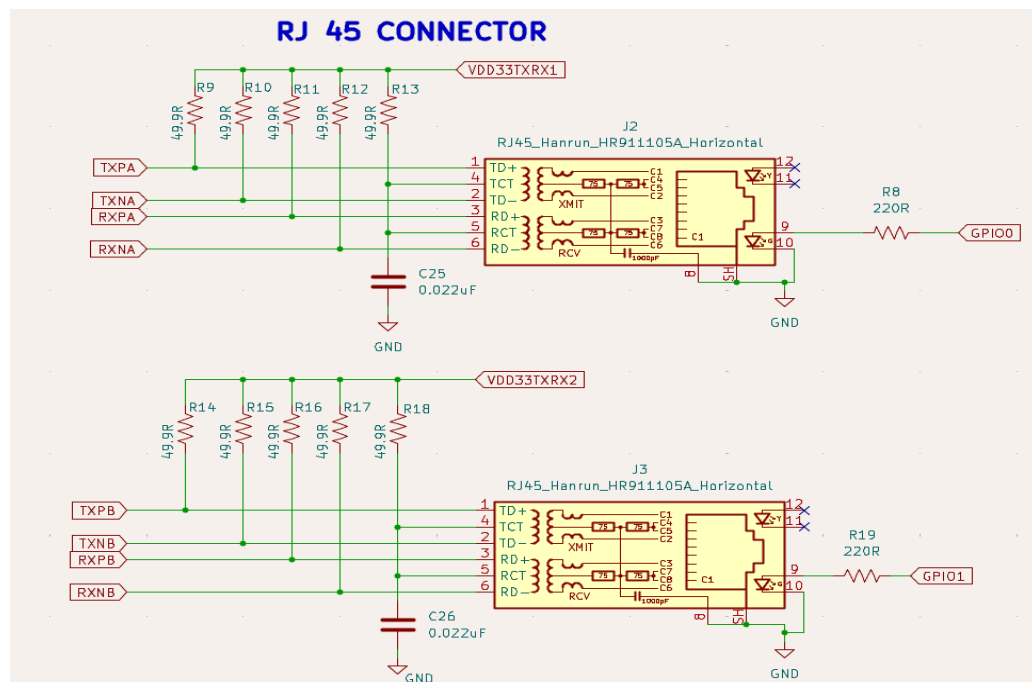


Figure (h)

The **RJ45 interface section** provides the physical connection between the LAN9252 EtherCAT controller and the external EtherCAT network. The design uses two RJ45 connectors, corresponding to the two EtherCAT ports, with each port carrying separate transmit and receive differential signals. The connectors include integrated Ethernet magnetics and status LED connections, while the surrounding termination and filtering components help provide the required electrical interface for reliable Ethernet communication.

Key Specifications:

- **Number of RJ45 ports:** 2



- **Interface:** 10/100 Mbps Ethernet / EtherCAT
- **Ethernet signals:** TXP, TXN, RXP and RXN for each port
- **Magnetics:** Integrated in the RJ45 connector
- **Termination resistors:** 49.9 Ω resistors used on the differential signal paths
- **Filtering:** 0.022 μ F capacitors provided in the interface section
- **Supply:** 3.3 V Ethernet interface supply
- **Status indication:** LED connections provided through GPIO signals
- **LED resistors:** 220 Ω current-limiting resistors

The two ports allow the board to provide the **EtherCAT network input and output connections**, enabling the device to be placed between EtherCAT network nodes. The termination resistors and integrated magnetics are important parts of the physical-layer interface and help maintain signal integrity during high-speed differential communication.



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PCB Layout

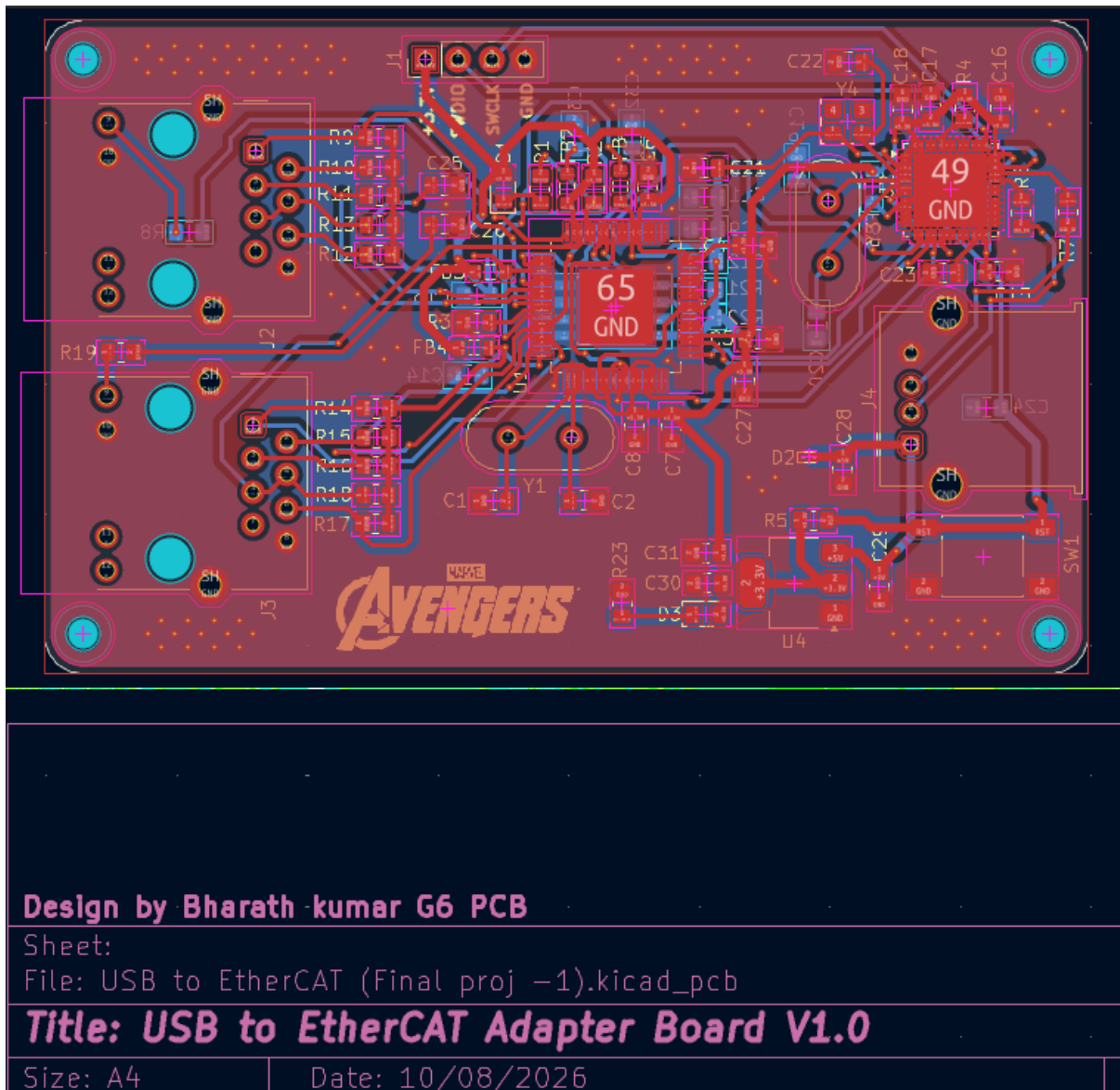


Figure (d) : Complete Layout

The completed PCB layout represents the physical implementation of the USB-to-EtherCAT adapter circuit developed from the schematic.



The layout also includes power and ground copper areas, mounting holes, board identification markings, and external connection points. The overall placement was organized to keep related components close to their respective ICs while maintaining a compact and practical board arrangement

Component Placement

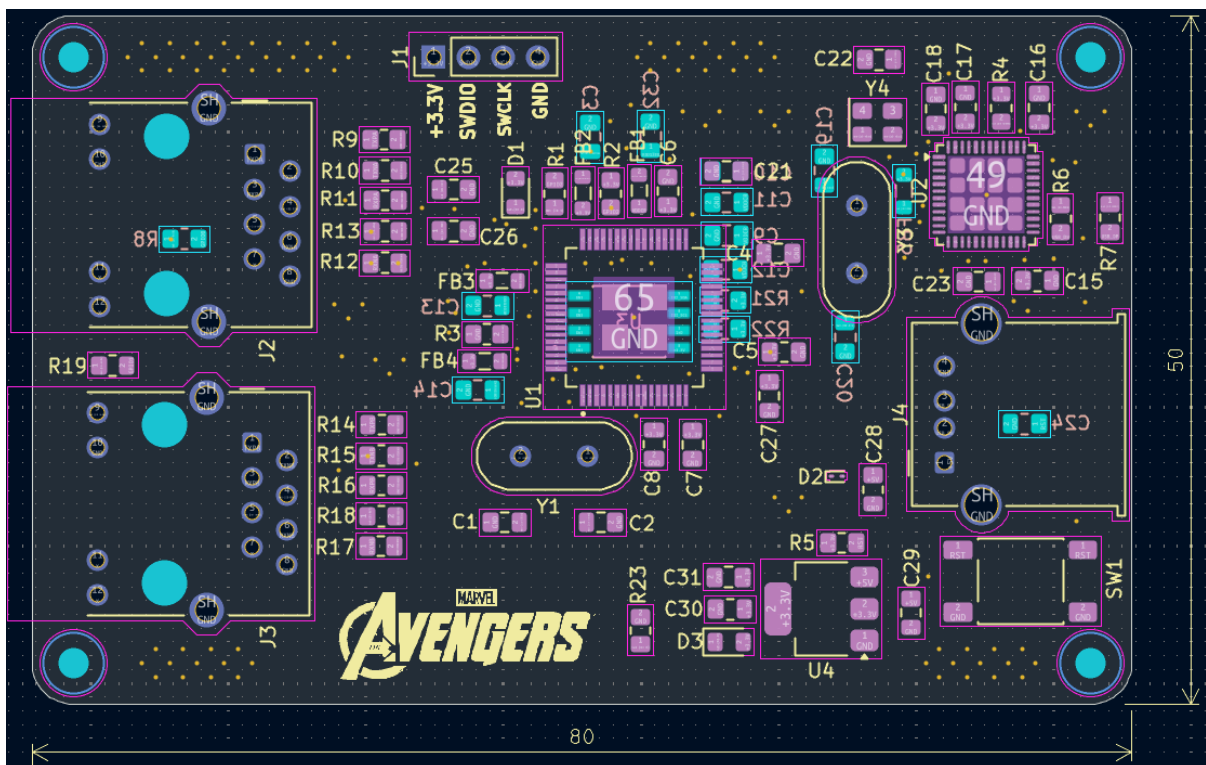


Figure (e): Front and Back Assembly View

Placement Considerations:

- **LAN9252 – Center:**
Placed near the center because it is the main EtherCAT controller and has connections to both the network interfaces and the STM32.
- **STM32F411CEUx – Upper-right:**
Placed separately from the LAN9252 to keep the microcontroller section organized. Its



supporting crystal, decoupling capacitors, reset components, and other passive components are positioned close to it.

- **RJ45 Connectors – Left side:**

The two RJ45 connectors are placed along the board edge because they are external network interfaces. Edge placement makes the connectors easily accessible.

- **EtherCAT supporting components – Near LAN9252/RJ45:**

The resistors, capacitors, and other passive components associated with the Ethernet interfaces are placed close to the corresponding signal paths to reduce unnecessary routing and maintain a compact layout.

- **Crystals – Close to respective ICs:**

The crystals are positioned close to the ICs they provide clock signals for. This helps keep clock connections short and reduces the possibility of unwanted noise or interference.

- **Decoupling capacitors – Close to IC power pins:**

The small capacitors are placed close to the power pins of the STM32 and LAN9252 to provide local supply filtering and reduce power-supply noise.

- **LDO regulator – Lower-right:**

The **AMS1117-3.3** power section is grouped separately from the main communication circuitry. This keeps the power-conversion components organized and helps separate the power section from sensitive signal circuitry.

- **SWD connector – Top:**

The programming/debug connector is positioned near the board edge so that it can be accessed easily when programming or debugging the STM32.

- **Mounting holes – Four corners:**

The mounting holes are placed at the four corners to provide mechanical support and allow the PCB to be securely mounted.

- **Board dimensions:**

The PCB shown has an approximate dimension of **80 mm × 50 mm**, providing a compact arrangement for the complete circuit.

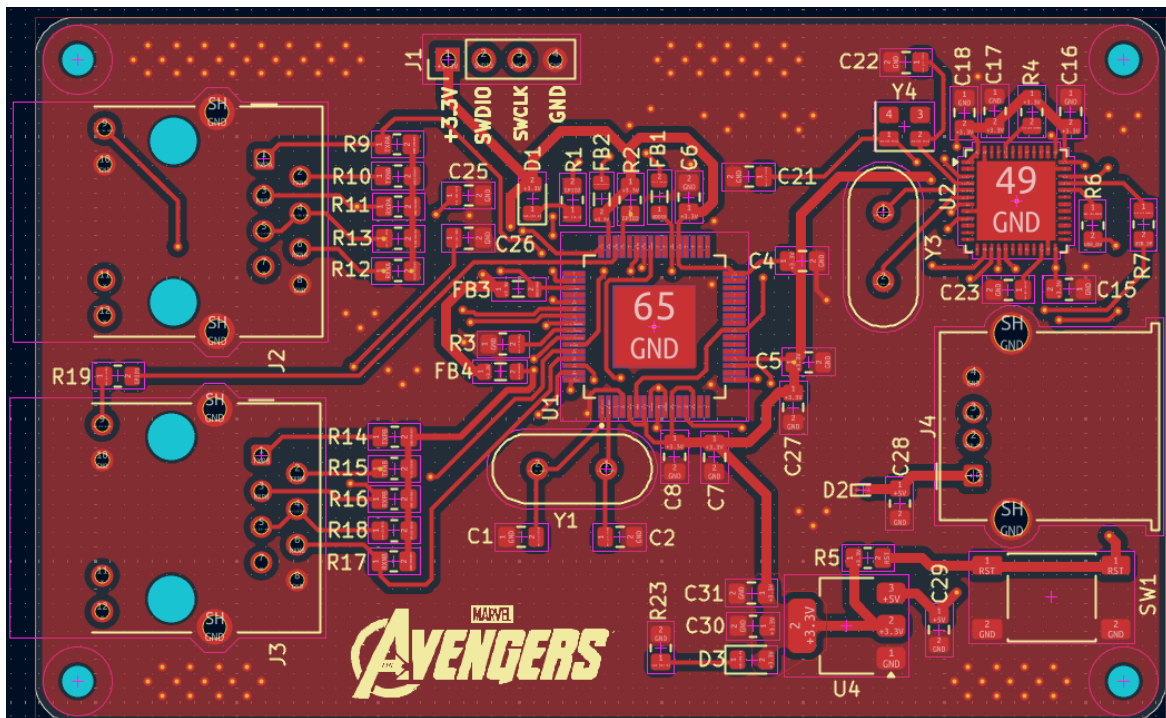


Figure (f): Front Copper Layer

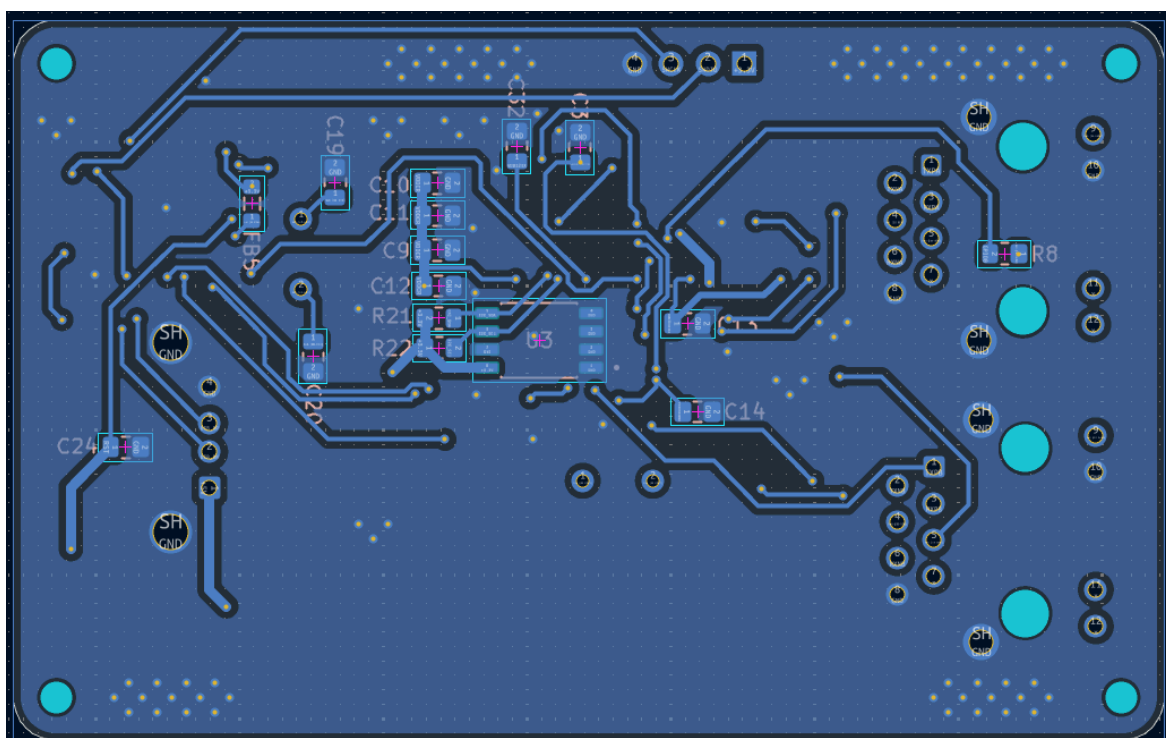


Figure (g): Back Copper Layer

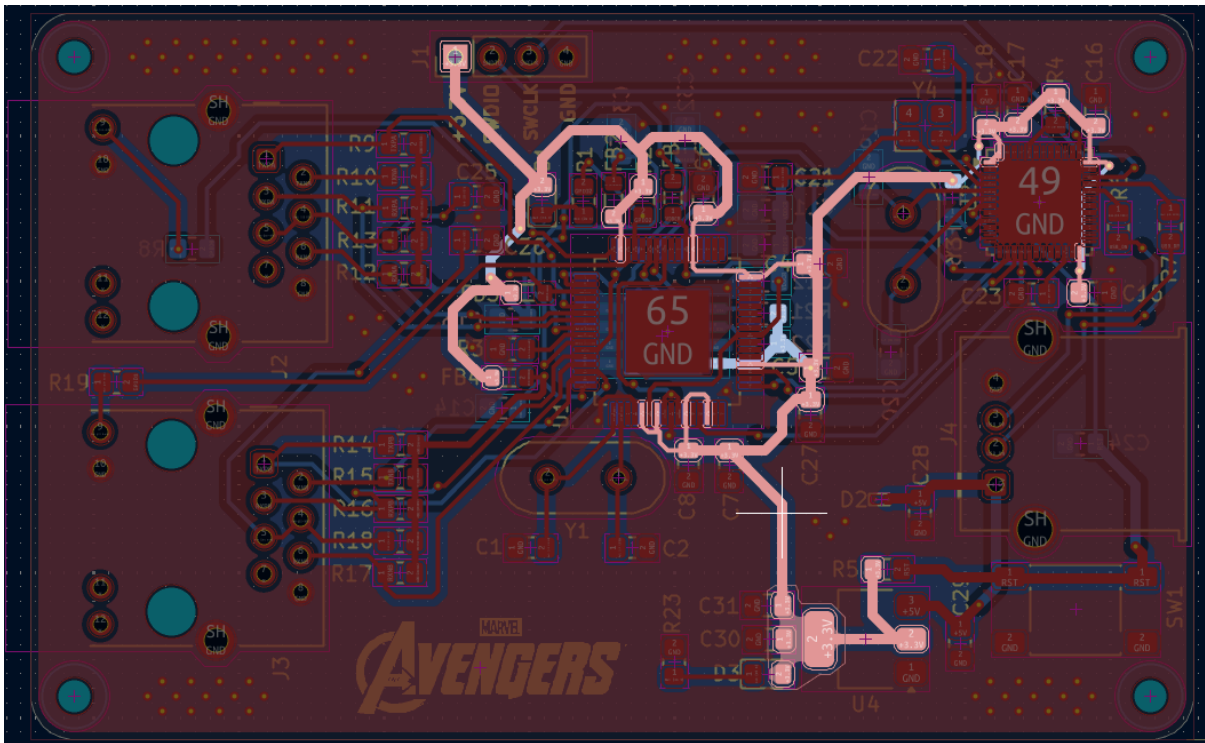


Figure (h): 3.3V Conducting Path

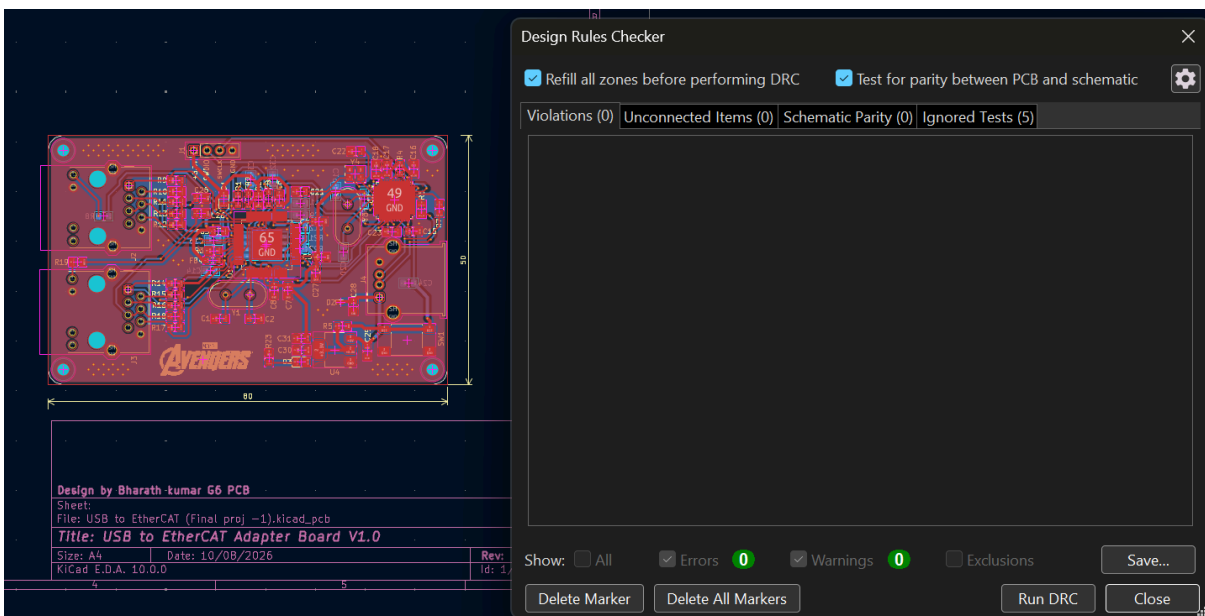


Figure (d): Verified design using Design Rule Checker(DRC)



Social/Industry Relevance

The **USB-to-EtherCAT Adapter Board** is useful in industrial automation, where different machines and devices need to communicate with each other quickly and reliably. EtherCAT is commonly used in areas such as robotics, CNC machines, motion control, and factory automation.

This project provides a way to connect a computer or embedded system through USB to an EtherCAT network. This can be useful for testing, monitoring, configuring, and communicating with EtherCAT-based devices without requiring a computer to have a built-in EtherCAT interface.

The project is also relevant from a learning and industry point of view because it involves **PCB design, USB communication, Ethernet communication, hardware interfacing, and embedded systems**. These are commonly used skills in industrial automation and electronics companies. Overall, the project gives practical experience in designing a hardware interface that can be used in real-world industrial communication systems.



Learnings and Reflections

This project gave me practical experience in taking a PCB design from the initial schematic to the final PCB layout. While working on the project, I gained a better understanding of how different functional blocks such as the STM32F411CEUx, LAN9252, USB interface, RJ45 interface, EEPROM, and power supply work together as a complete system. It also helped me understand that PCB design involves more than just making electrical connections, as component placement, routing, grounding, and power distribution all need to be considered.

Key Learnings:

- Gained hands-on experience with KiCad schematic and PCB design.
- Learned how to use component datasheets while making design decisions.
- Understood how to plan component placement before routing.
- Gained practical knowledge of PCB routing, vias, grounding, and power distribution.
- Understood the importance of placing decoupling capacitors and clock components close to the relevant ICs.
- Learned how different interfaces such as USB, SPI, I²C, and Ethernet/EtherCAT are brought together on one PCB.
- Gained experience with ERC, DRC, and 3D PCB visualization.

Reflection:

One of the main things I learned from this project is that a good PCB design requires both circuit-level understanding and careful physical planning. Small decisions, such as where to place a component or how to route a connection, can affect the overall organization and performance of the board. Working through the design process also made it easier to understand how the concepts learned in electronics are applied in an actual PCB development workflow.

Conclusion and Future Scope



Conclusion

The project involved the complete design of a USB-to-EtherCAT Adapter Board V1.0, starting from the schematic and progressing through component selection, PCB placement, routing, and design verification. The final design brings together the STM32F411CEUx, LAN9252 EtherCAT controller, USB interface, dual RJ45 interfaces, I²C EEPROM, and 5 V to 3.3 V power supply on a single PCB. Working through these stages provided practical experience in PCB development and helped in understanding the design considerations involved in building a communication-oriented board.

Future Scope

The designed PCB can be further developed from a design-stage prototype into a **fabricated and tested hardware board**. Future work can include manufacturing the PCB, assembling the components, and performing electrical and functional testing of the USB and EtherCAT interfaces.

Possible future improvements include:

- PCB fabrication and hardware testing to verify the complete design.
- Testing USB-to-EtherCAT communication with a host computer and EtherCAT devices.
- Performing signal-integrity and power-supply measurements on the fabricated board.
- Optimizing the PCB layout based on practical testing results.
- Developing the required firmware/software for complete USB-to-EtherCAT operation.

Project Files

The complete project files, including the schematic, PCB layout, are provided in the following Google Drive folder.



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Google Drive: [\[Project Files – USB-to-EtherCAT Adapter Board V1.0\]](#)

Previous PCB Designs

S.No	Title of Project	Design Files Link
1	LED Design	LINK
2	LDR Design	LINK
3	DC to DC Converter	LINK
4	USB to TTL Converter	LINK
5	TSH82 Op Amp	LINK
6	8051 board	LINK
7	Arduino Pro Mini	LINK
8	Arduino 2.0	LINK
9	Arduino 3.0	LINK
10	STM32 board	LINK